K-Means Clustering — MCQs

1. What is the primary objective of the K-Means clustering algorithm?

A. Maximize classification accuracy
B. Minimize the within-cluster sum of squared distances
C. Maximize the distance between all data points
D. Minimize the number of clusters

2. Which type of learning does K-Means represent?

A. Supervised learning
B. Semi-supervised learning
C. Unsupervised learning
D. Reinforcement learning

3. In K-Means, what does the parameter K represent?

A. Number of features
B. Number of iterations
C. Number of clusters
D. Number of observations

4. Which of the following is the usual sequence of steps in K-Means?

A. Initialize centroids → Assign points → Update centroids → Repeat
B. Assign points → Initialize centroids → Update centroids → Stop
C. Initialize centroids → Update centroids → Assign points → Stop
D. Update centroids → Initialize centroids → Assign points → Repea

5. In the standard K-Means algorithm, a data point is assigned to:

A. The cluster with the largest centroid
B. The nearest centroid
C. The farthest centroid
D. A randomly selected cluster

6. Which distance metric is most commonly associated with standard K-Means?

A. Manhattan distance
B. Euclidean distance
C. Hamming distance
D. Jaccard distance

7. What happens to a centroid during the update step of K-Means?

A. It is moved to the median of the cluster
B. It is moved to the mean of the points assigned to that cluster
C. It is moved to the farthest point
D. It remains fixed

8. Consider the one-dimensional points {2, 4, 6} assigned to a single cluster. What will be the new centroid?

A. 2
B. 4

C. 6
D. 12

9. Suppose a cluster contains points (1,2), (3,4), and (5,6). What is its centroid?

A. (2,3)
B. (3,4)
C. (4,5)
D. (9,12)

10. Which of the following can cause K-Means to produce different results when run multiple times?

A. Random initialization of centroids
B. Number of features
C. Dataset size
D. Distance calculation

11. Why is feature scaling often important before applying K-Means?

A. K-Means cannot handle numerical data
B. Features with larger scales can dominate the distance calculation
C. Scaling automatically determines the value of K
D. Scaling eliminates the need for centroids

12. Consider two points P(2,3) and Q(5,7). What is their Euclidean distance?

A. 4
B. 5
C. 6
D. 7

13. If the value of K is increased from 3 to 5, what generally happens to the K-Means objective function (within-cluster SSE)?

A. It always increases
B. It generally decreases or remains the same
C. It becomes exactly zero
D. It becomes undefined

14. Which method is commonly used to help determine an appropriate value of K?

A. Elbow method
B. Gradient descent
C. Backpropagation
D. ROC curve

15. In the Elbow Method, what is plotted against K?

A. Accuracy
B. Precision
C. Within-cluster sum of squared errors
D. Number of features

16. A major limitation of K-Means is that it:

A. Cannot work with numerical data
B. Requires the number of clusters to be specified in advance
C. Cannot calculate distances
D. Always produces the global optimum

17. Which type of cluster structure is K-Means generally most suitable for?

A. Approximately spherical and similarly sized clusters
B. Arbitrarily shaped clusters
C. Nested hierarchical clusters
D. Highly irregular clusters with varying densities

18. Suppose K-Means converges to a solution. Which statement is TRUE?

A. It is guaranteed to be the global optimum
B. It has reached a stable solution under the chosen initialization
C. All data points must be identical
D. The number of clusters automatically becomes optimal

19. Why can K-Means be sensitive to outliers?

A. Centroids are calculated using the mean
B. K-Means uses categorical variables
C. Outliers are always removed
D. K-Means does not use distance

20. Suppose two centroids are C_1=(1,1) and C_2=(10,10), and a point is P=(2,3). To which cluster will P be assigned?

A. Cluster 1
B. Cluster 2
C. Both clusters
D. Cannot be determined